technische universität dortmund

Chair for Reliability Engineering

PFEM Benchmark Catalogue

From Deterministic Sweeps to Probabilistic Analysis

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Where We Are

Done

- ✓ **Foundation:** all 87 PFEM cases build, run and verified on Linux
- ✓ **Deterministic sweeps** + 3D post-processing (demo today: p6.12 slope)
- ✓ **Phase 1:** universal Quantity-of-Interest extraction + Monte Carlo
- ✓ **Phase 2:** Latin Hypercube, correlated inputs, sensitivity tornado

Next

- ▶ Phase 3: pluggable runner for non-PFEM codes
- ▶ Phase 4: mesh-refinement convergence studies
- ▶ Regression testing vs reference outputs

Handover: complete documentation set and a 3-minute reproducible test, all on GitHub (final slides).

The story: we went from *one number per run* to *full probabilistic and sensitivity analysis* across

What We Built

PFEM Benchmark Catalogue

- ✓ 87 Fortran programs (Chapters 4–11) running on Linux
- ✓ YAML metadata auto-generated from Fortran source
- ✓ MATLAB sweep GUI: pick a case, vary parameters, view results

How It Works

1. YAML defines tunable parameters per program
2. MATLAB injects values into `.dat` input files
3. Fortran binary runs in an isolated sandbox
4. Outputs (`.res`, `.ensi`) parsed and plotted automatically

Demo: 3D slope stability analysis (p6.12) — Mohr-Coulomb with strength reduction

p6.12 — 3D Slope Stability (Mohr-Coulomb)

Physical Problem

- 3D embankment slope, elastic-perfectly plastic
- Mohr-Coulomb failure criterion with 5 material zones
- Hex8 mesh, 20-point Gaussian integration

Strength Reduction Factor (SRF)

Shear strength is scaled down uniformly at each step:

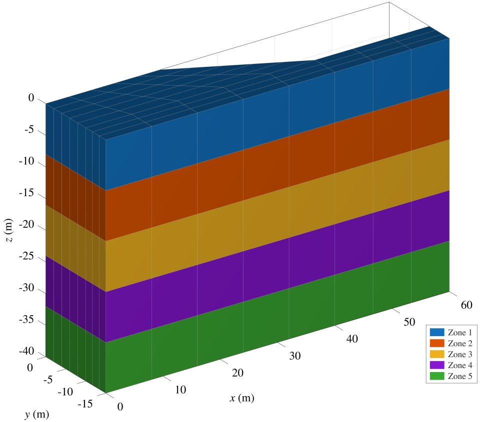
$$\hat{c} = \frac{c}{\mathrm{SRF}}, \qquad \tan\hat{\phi} = \frac{\tan\phi}{\mathrm{SRF}}$$

Failure = last SRF at which Newton-Raphson

Default Parameters

Parameter	Value
Cohesion c	60 kPa
Friction ϕ	20°
Dilation ψ	0°
Young's E	10^5 Pa
Poisson ν	0.3
SRF range	1.0 – 1.60
Iteration limit	1000

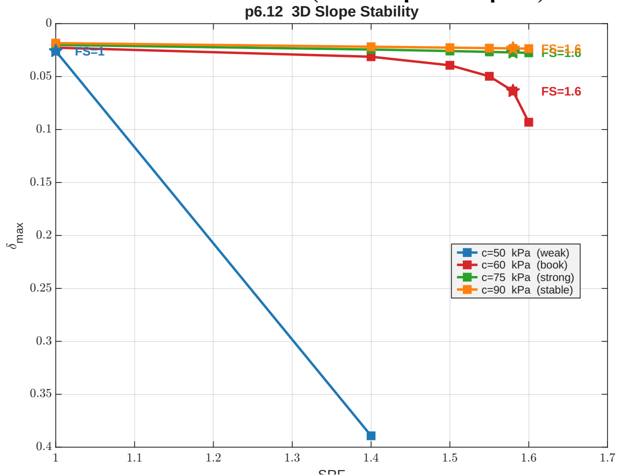
p6.12 — Material Zones



5 horizontal soil layers

- Each zone has its own cohesion c and friction angle ϕ
- Geometry is **identical** across all sweep scenarios
- Only the base cohesion value changes; zone proportions are preserved
- Slope face visible on the right side

p6.12 — SRF Curves (Sweep Output)

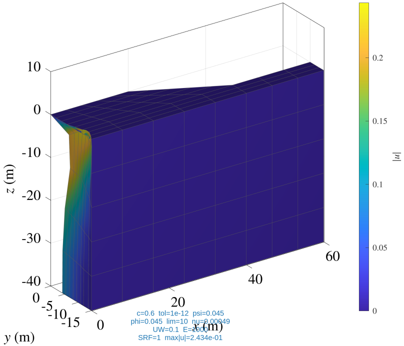


What this shows

- X-axis: Strength Reduction Factor applied to c and ϕ
- Y-axis: maximum displacement $\delta_{\max}$
- $c = 50$ kPa: slope fails at SRF$\approx$1.4 — displacement diverges
- $c = 60$ kPa (book): FS$\approx$1.6
- $c = 75$, $c = 90$: stable — no divergence

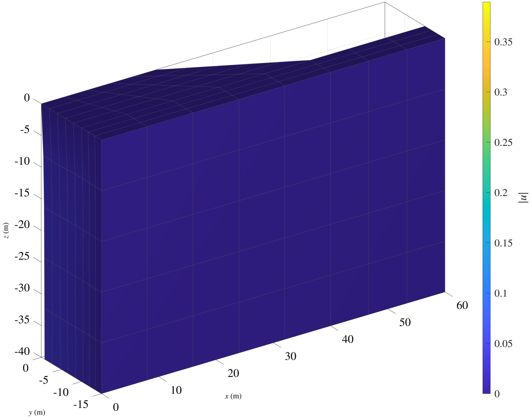
Key: diverging $\delta_{\max}$ signals slope failure.

p6.12 — 3D Deformed Mesh at Failure (All Scenarios)



- Each panel: deformed mesh at the highest converged SRF step; colour = $|u|$
- Top-left (c=50): severe slope failure. Bottom-right (c=90): minimal deformation

p6.12 — $c = 50\,\mathrm{kPa}$: Slope Failure

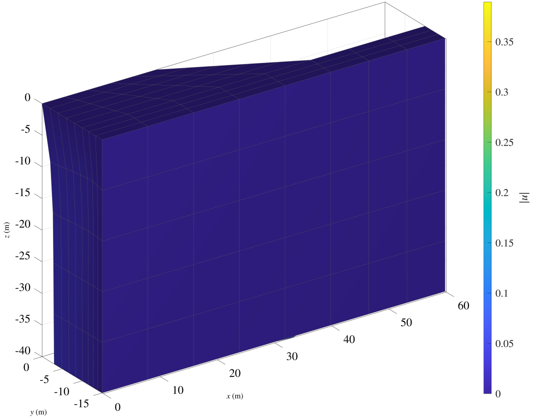


Weak soil scenario

- Newton-Raphson diverges at SRF ≈ 1.4
- Clear circular slip surface visible at the slope crest
- Large horizontal displacement at the toe
- Colourmap confirms displacement localisation along failure plane

This scenario demonstrates **how the tool detects failure** automatically.

p6.12 — $c = 60\,\text{kPa}$: Book Default



Reference scenario

- Incipient failure at SRF ≈ 1.6 (iteration limit reached)
- Deformed shape matches book Fig. 6.55 (PFEM 5th ed.)
- Moderate displacement — slope is on the verge of collapse

Validates our pipeline against the published reference solution.

p6.12 — Verification vs. Book (Figs. 6.54–6.55)

SRF vs. Max Displacement (c=60, ϕ=20°)

SRF	δ_{max}	Iterations
1.00	0.02266	16
1.40	0.03123	77
1.50	0.03930	208
1.55	0.04971	380
1.58	0.06363	703
1.60	0.09308	1000

Failure at SRF ≈ 1.58–1.60

Verification

- ✓ SRF–displacement curve matches book Fig. 6.54
- ✓ Failure SRF ≈ 1.58–1.60 agrees with reference
- ✓ 3D slip surface visible in deformed mesh (Fig. 6.55)
- ✓ Iteration count growth consistent with limit-state analysis

Phase 1 — From One Answer to a Distribution

The problem

- A deterministic run gives *one* number for *one* set of inputs
- Real material properties are uncertain (scatter, test variability)
- The old extractor only understood slope-stability output

What was added

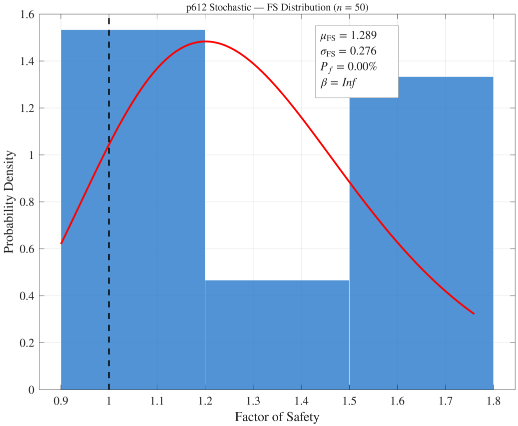
✓ A **universal QoI extractor**: pulls the right physical answer for **all 87** cases (8 case types)

Case type → Quantity of Interest

Case type	QoI
Slope stability	Factor of Safety
Plasticity	Limit load
Elastic	Max displacement
Seepage	Max head
Consolidation	Degree of consol.
Eigenvalue	ω^2 (and f_1)
Dynamics	Peak displacement
Thermal	Max temperature

87 / 87 cases return a meaningful QoI from their default run.

Phase 1 — Result: Slope Reliability (p6.12)

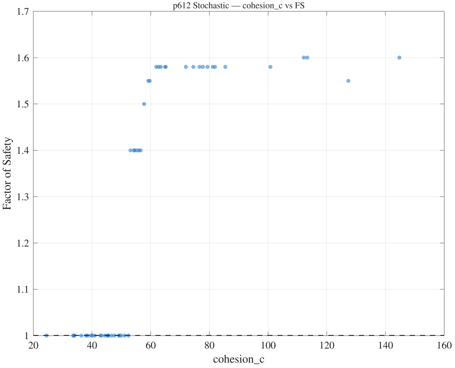


Cohesion sampled as lognormal

- Each Monte Carlo sample runs the full 3D Fortran solver
- Output: a **distribution** of Factor of Safety, not a single value
- Gives **probability of failure** $P(\text{FS} < 1)$ and reliability index β

This is the language a geotechnical reviewer actually wants: not "FS = 1.58" but "probability of failure = X%".

Phase 2 — Latin Hypercube: Same Cost, Better Coverage



Variance reduction vs plain Monte Carlo

Samples	LHS std	IID std
25	0.75	4.92 (**6.5**×)
50	0.54	3.42
100	0.20	2.33
200	0.11	1.58 (**14**×)

- Same number of expensive Fortran runs
- 6–14× tighter estimate of the mean
- Scatter (left) shows FS rising smoothly with cohesion — physically

Phase 2 — Correlated Inputs (Iman–Conover)

Why

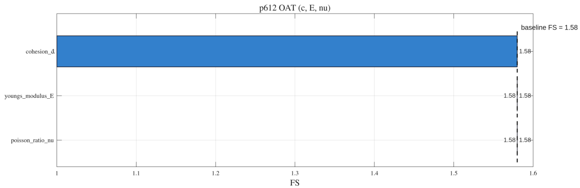
- In real soils, cohesion c and friction ϕ are correlated (typically negatively)
- Sampling them independently is unphysical
- Iman–Conover induces a **target correlation** while keeping each input's distribution exact

Reproduced target correlations (n=500)

Target ρ	Observed ρ
-0.70	-0.654
-0.30	-0.275
$+0.00$	-0.002
$+0.50$	$+0.494$

Targets in $[-0.7, +0.5]$ reproduced within 5%, marginals preserved exactly.

Phase 2 — Sensitivity: Which Input Matters?



One-at-a-time tornado (p6.12)

- Each input varied $\pm 1\sigma$; bar length = effect on FS
- **Cohesion** dominates (spread 0.58)
- E **and** ν have *zero* effect on FS

Exactly the textbook expectation: in strength-reduction analysis only strength parameters move the safety factor; stiffness only affects displacements. The tool **recovers the physics**.

Why We Trust It — Validation

Case	Computed	Reference	Match
p6.12 slope FS (c=60, ϕ=0)	1.58	Book Fig. 6.54	✓
p6.1 limit load (σ_y=100)	515	Prandtl $(2+\pi)\sigma_y = 514$	✓ 0.2%
p10.1 beam ω^2	0.00382	Analytic cantilever 0.00402	✓
LHS variance reduction	6.5–14×	McKay et al. (1979)	✓
Correlation reproduction	within 5%	Iman–Conover (1982)	✓

- Results match the **textbook** and **analytical** solutions, not just "it runs"
- A single 3-minute script (`test_phase2_multi_case.m`) reproduces this evidence on demand

Handover — Anyone Can Pick This Up

Everything is documented and on GitHub

- `ARCHITECTURE.md` — end-to-end flow, how every file connects, the call graph, every problem solved
- `HANDOVER.md` — fresh-machine setup, status, gotchas
- `GUIDE.md` / `matlab/README.md` — usage and function reference
- `PROGRESS.md` — this validation evidence in full

Reproducible from a cold start

- A literal "**zero to first result**" runbook: clone $\rightarrow$ build $\rightarrow$ run $\rightarrow$ figure
- 3-minute test script reproduces all validation
- Documented patches make the textbook Fortran compile on Linux

The only thing not in the repo is the licensed PFEM source — but the exact patches to rebuild it are.

Conclusions & Next Steps

What was achieved

- ✓ 87 PFEM benchmarks verified on Linux (token-based YAML, zero source changes)
- ✓ Universal QoI extraction across all 8 case types
- ✓ Monte Carlo + Latin Hypercube + correlated inputs
- ✓ One-at-a-time sensitivity (tornado) recovering the physics
- ✓ Validated vs textbook and analytical

Next steps

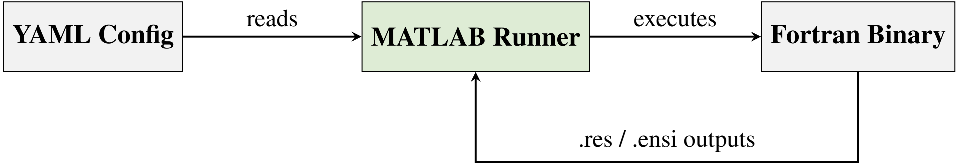
- ► Phase 3: pluggable runner for non-PFEM codes
- ► Phase 4: mesh-resolution convergence sweeps (*nxe*, *nye*)
- ► Automated regression testing against reference outputs
- ► User-selectable probe node for boundary-bound QoIs

Thank You!

Questions?

Backup Slides

Pipeline Architecture



YAML acts as the bridge between rigid Fortran and flexible MATLAB.

Per-run steps

1. Locate binary + template `.dat`
2. Copy to an isolated sandbox folder
3. Inject override values as positional tokens
4. Execute; parse output files

Example tunable tokens

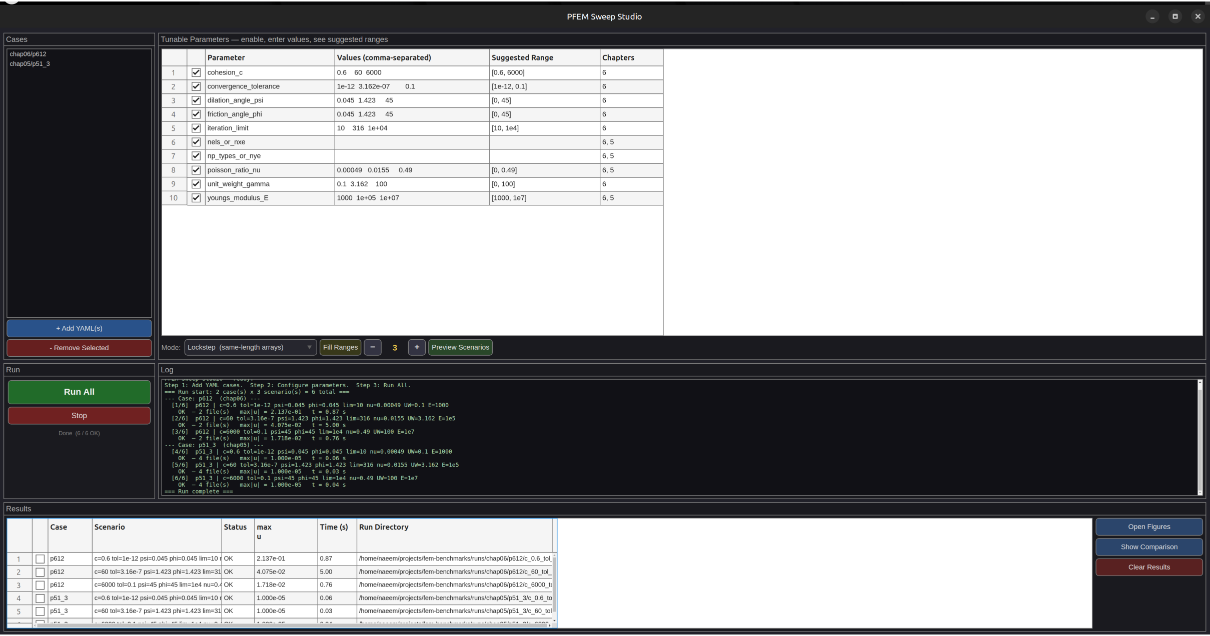
- `youngs_modulus_E` → replaces value in `.dat`
- `cohesion_c`, `friction_phi` → MC sweep
- One YAML, many scenarios — no

Sweep GUI — `pfem_sweep_gui`

A single MATLAB interface for all 87 benchmark cases:

- Browse cases by chapter; inspect tunables with suggested ranges
- Define sweeps on a **log or linear** grid
- Execute variants in isolated sandboxes
- Auto-generate summary figures (mesh, displacement, SRF, 3D EnSight)

1. Select Case
↓
2. Browse Tunables
↓
3. Set Sweep Grid
↓
4. Run All Variants
↓
5. View Summary Plots


PFEM Sweep Studio
Cases
chap06/p612
chap05/p51_3
+ Add YAML(s)
- Remove Selected
Tunable Parameters — enable, enter values, see suggested ranges
Parameter | Values (comma-separated) | Suggested Range | Chapters
1 cohesion_c | 0.6 60 6000 | [0.6, 6000] | 6
2 convergence_tolerance | 1e-12 3.162e-07 0.1 | [1e-12, 0.1] | 6
3 dilation_angle_psi | 0.045 1.423 45 | [0, 45] | 6
4 friction_angle_phi | 0.045 1.423 45 | [0, 45] | 6
5 iteration_limit | 10 316 1e+04 | [10, 1e4] | 6
6 nels_or_nxe | | | 6, 5
7 np_types_or_nye | | | 6, 5
8 poisson_ratio_nu | 0.00049 0.0155 0.49 | [0, 0.49] | 6, 5
9 unit_weight_gamma | 0.1 3.162 100 | [0, 100] | 6
10 youngs_modulus_E | 1000 1e+05 1e+07 | [1000, 1e7] | 6, 5
Mode: Lockstep (same-length arrays)
Fill Ranges
−
3
+
Preview Scenarios
Run
Run All
Stop
Done (6 / 6 OK)
Log
Step 1: Add YAML cases. Step 2: Configure parameters. Step 3: Run All.
=== Run start: 2 case(s) x 3 scenario(s) = 6 total ===
--- Case: p612 (chap06) ---
[1/6] p612 | c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10 nu=0.00049 UW=0.1 E=1000
OK – 2 file(s) max|u| = 2.137e-01 t = 0.87 s
[2/6] p612 | c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=316 nu=0.0155 UW=3.162 E=1e5
OK – 2 file(s) max|u| = 4.075e-02 t = 5.00 s
[3/6] p612 | c=6000 tol=0.1 psi=45 phi=45 lim=1e4 nu=0.49 UW=100 E=1e7
OK – 2 file(s) max|u| = 1.718e-02 t = 0.76 s
--- Case: p51_3 (chap05) ---
[4/6] p51_3 | c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10 nu=0.00049 UW=0.1 E=1000
OK – 4 file(s) max|u| = 1.000e-05 t = 0.06 s
[5/6] p51_3 | c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=316 nu=0.0155 UW=3.162 E=1e5
OK – 4 file(s) max|u| = 1.000e-05 t = 0.03 s
[6/6] p51_3 | c=6000 tol=0.1 psi=45 phi=45 lim=1e4 nu=0.49 UW=100 E=1e7
OK – 4 file(s) max|u| = 1.000e-05 t = 0.04 s
=== Run complete ===
Results
Case | Scenario | Status | max u | Time (s) | Run Directory
1 p612 | c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10 r | OK | 2.137e-01 | 0.87 | /home/naeem/projects/fem-benchmarks/runs/chap06/p612/c_0.6_tol_
2 p612 | c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=31 | OK | 4.075e-02 | 5.00 | /home/naeem/projects/fem-benchmarks/runs/chap06/p612/c_60_tol_
3 p612 | c=6000 tol=0.1 psi=45 phi=45 lim=1e4 nu=0.4 | OK | 1.718e-02 | 0.76 | /home/naeem/projects/fem-benchmarks/runs/chap06/p612/c_6000_tc
4 p51_3 | c=0.6 tol=1e-12 psi=0.045 phi=0.045 lim=10 r | OK | 1.000e-05 | 0.06 | /home/naeem/projects/fem-benchmarks/runs/chap05/p51_3/c_0.6_to
5 p51_3 | c=60 tol=3.16e-7 psi=1.423 phi=1.423 lim=31 | OK | 1.000e-05 | 0.03 | /home/naeem/projects/fem-benchmarks/runs/chap05/p51_3/c_60_tol
Open Figures
Show Comparison
Clear Results

Problem Statement

Developing benchmark Fortran codes (PFEM, 5th ed.) by automating YAML metadata generation to tokenise `.dat` inputs.

Key Challenges

- **Legacy Input:** Unstructured `.dat` files.
- **Opaque Parsing:** The `read10` subroutine hides logic.
- **Constraint:** Source code must remain untouched.

Objectives

- **Reproducibility:** Run any case reliably.
- **Interoperability:** Drive execution via MATLAB.
- **Searchability:** Consistent YAML metadata.

Path to Completion

To transition from prototype to a **research-ready benchmark repository**:

1. **Standardise YAML Schema (All Chapters)**
   - Uniform metadata keys and consistent tunable definitions.
2. **Define Scope of Tunables**
   - **Phase 1:** Material props (E, ν, k), loads.
   - **Phase 2:** Mesh resolution (*nxe*, *nye*).
3. **Implement Validation Layer**
   - Health checks and regression testing against reference outputs.

p5.1 — Problem Setup (Linear Elasticity 2D)

Physical Problem

- Plane-stress elastic solid under point loads
- Mesh: 3×2 bilinear quadrilateral elements (12 nodes, 6 elements)
- Fixed bottom edge; two point loads applied at top

Tunable Parameters

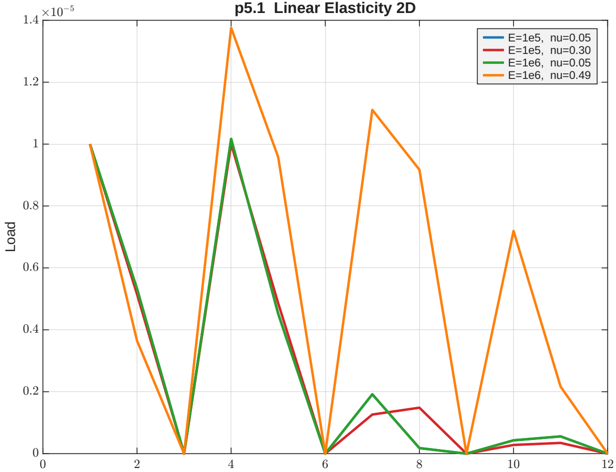
Parameter	Symbol	Range
Young's modulus	E	10^4–10^8
Poisson's ratio	ν	5×10^{-4}–0.49

YAML snippet


```yaml
id: pfem5_ch05_p51_p51_3
title: PFEM p51 -- case p51_3
purpose: Linear Elasticity 2D

tunable_parameters:
- name: youngs_modulus_E
  current_value: '1.0e6'
  suggested_range: [1e4, 1e9]
- name: poisson_ratio_nu
  current_value: '0.3'
  suggested_range: [0.0, 0.49]
```

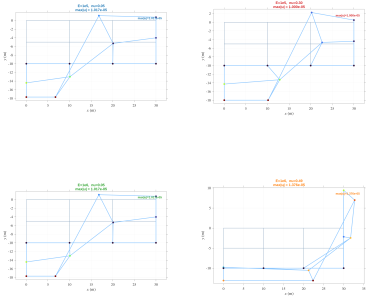
p5.1 — Load–Displacement Overlay



What this shows

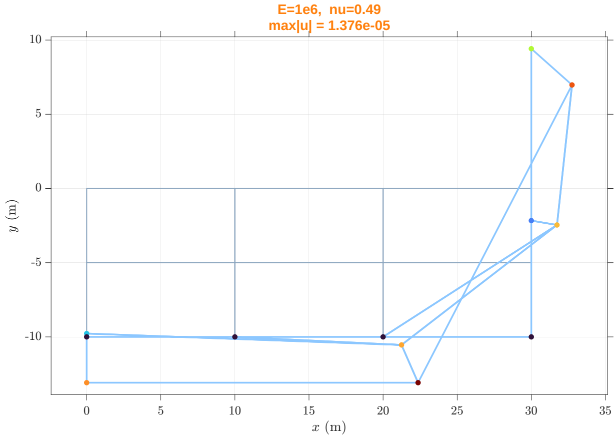
- 4 scenarios: E and ν varied on a log grid
- ν=0.49: near-incompressible, strong Poisson effect
- Low ν scenarios cluster together — stiffness dominates

p5.1 — Deformed Shapes (All Scenarios)



- Poisson ratio dominates the deformed shape; $\nu = 0.49$ shows strong lateral bulging
- Grey = undeformed mesh, blue = amplified deformed shape

p5.1 — Zoom: $E=10^6$, $\nu=0.49$ (Near-Incompressible)



Observation

- Near-incompressible material ($\nu\to0.5$)
- Large lateral expansion dominates the deformed shape
- Vertical load produces significant horizontal displacement

p5.1 — Verification vs. Book (Table 5.3)

Nodal Displacements

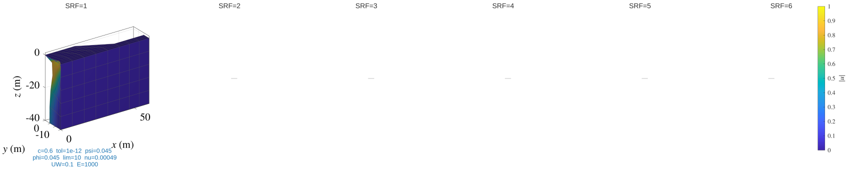
Default: $E = 10^6$, $\nu = 0.3$, plane stress

Node	u_x	u_y
1	0.000	-1.000×10^{-5}
4	9.446×10^{-6}	-1.000×10^{-5}
5	8.592×10^{-6}	-4.235×10^{-6}
10	0.000	$+7.196 \times 10^{-6}$

Verification

- ✓ Displacements match book Table 5.3 exactly
- ✓ Integration point stresses match Table 5.3
- ✓ Sweep shows correct $1/E$ scaling

p6.12 — SRF Progression (Deformation Sequence)



- Rows = scenarios (c=50 → 90 kPa). Columns = SRF steps 1.0–1.6
- Weak soil: progressive failure visible across columns. Strong soil: stable throughout